

# SHI-XIN ZHANG (张士欣)

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*Associate Professor at IOP CAS and former Senior Research Scientist at Tencent. Pioneering the intersection of Quantum Computing and Artificial Intelligence. Creator of TensorCircuit, a widely used open-source quantum framework. Distinguished for bridging fundamental physics insights with robust engineering practices to deliver advanced tools and solutions, backed by 40+ patents and 65+ publications.*

## EXPERIENCE

**Institute of Physics, Chinese Academy of Sciences** Associate Professor 2024 – Present

- Established and lead a research group in quantum physics, quantum computing, and AI. Mentor PhD students, postdoctoral researchers, and undergraduates on projects connecting deep-learning infrastructure, quantum many-body physics, and scientific computing. Secured competitive research grants totaling over RMB 15 million from NSFC, CAS, and the National Science and Technology Major Project.

**Tencent Quantum Laboratory** Senior Research Scientist 2021 – 2024

- Led a multidisciplinary research team on the analysis and design of near-term quantum algorithms, quantum simulation schemes, and quantum–AI hybrid methods, and collaborated with partners in finance, biology, energy, materials, and IT to explore industry applications with potential quantum advantage. Created and developed TensorCircuit, a high-performance, full-featured open-source quantum framework with an advanced tensor-network engine and native integration with modern machine-learning frameworks; led its research, architecture, core development, and open-source maintenance.

## EDUCATION

**Tsinghua University** PhD in Physics 2016 – 2021

*Institute for Advanced Study* Advisor: Prof. Hong Yao

PhD Thesis: Differentiable Programming in Quantum Physics

Outstanding PhD Award (87 amongst 2981) / Outstanding PhD thesis at Tsinghua University

**Tsinghua University** BSc in Physics 2012 – 2016

*Department of Physics* GPA: 95/100 Rank: 1/95

**Top 1** in 2012 National College Entrance Examination in Hebei Province

## RESEARCH

### Interests

- Quantum computing: quantum algorithm, quantum simulation, quantum noise, quantum software, quantum information.
- Artificial intelligence: the interplay between machine learning infrastructure/models/philosophy with quantum many-body physics, quantum machine learning.
- Non-equilibrium physics: novel phases and phase transitions in non-equilibrium quantum systems including quantum chaos, many-body localization, time crystal, (monitored) random quantum circuits, etc.

### Publications

65 publications and preprints in total (54 as the first or corresponding author), 49 published in peer-reviewed journals (41 as the first or corresponding author), including 9 in *Physical Review Letters*, 1 in *Nature Communications*, and 1 in *PRX Quantum*. 2,000+ citations; h-index 23. Selected works as the **first**, **joint first**<sup>†</sup> and **corresponding**\* author are listed below; please see my Google Scholar profile for the full publication list.

- Yu-Qin Chen\*, and **Shi-Xin Zhang**\*, *Intrinsic preservation of plasticity in continual quantum learning*, *PRX Quantum* **7**, 033003 (2025).
- Chun-Yue Zhang, Zi-Xiang Li\*, and **Shi-Xin Zhang**\*, *Entanglement Growth from Entangled States: A Unified Perspective on Entanglement Generation and Transport*, *Physical Review Letters* **137**, 020404 (2026).

3. Wei-Xuan Chang, Shuai Yin\*, **Shi-Xin Zhang\***, and Zi-Xiang Li\*, *Imaginary-time Mpemba effect in quantum many-body systems*, Physical Review Letters **136**, 100403 (2026).
4. Shuo Liu<sup>†</sup>, Hao-Kai Zhang<sup>†</sup>, Shuai Yin, and **Shi-Xin Zhang\***, *Symmetry restoration and quantum Mpemba effect in symmetric random circuits*, Physical Review Letters **133**, 140405 (2024).
5. Shuo Liu, Ming-Rui Li, **Shi-Xin Zhang\***, and Shao-Kai Jian\*, *Entanglement structure and information protection in noisy hybrid quantum circuits*, Physical Review Letters, **132**, 240402 (2024).
6. Shuo Liu, **Shi-Xin Zhang\***, Chang-Yu Hsieh, Shengyu Zhang, and Hong Yao\*, *Discrete time crystal enabled by Stark many-body localization*, Physical Review Letters **130**, 120403 (2023).
7. **Shi-Xin Zhang**<sup>†</sup>, Zhou-Quan Wan<sup>†</sup>, Chee-Kong Lee, Chang-Yu Hsieh\*, Shengyu Zhang, and Hong Yao\*, *Variational quantum-neural hybrid eigensolver*, Physical Review Letters **128**, 120502 (2022).
8. **Shi-Xin Zhang** and Hong Yao\*, *Universal properties of many-body localization transitions in quasiperiodic systems*, Physical Review Letters **121**, 206601 (2018).
9. Yu-Qin Chen\* and **Shi-Xin Zhang\***, *Superior resilience to poisoning and amenability to unlearning in quantum machine learning*, Nature Communications **17**, 3716 (2026).
10. Hui Yu, Jiangping Hu\*, and **Shi-Xin Zhang\***, *Hilbert subspace imprint: a new mechanism for non-thermalization*, Chinese Physics Letters **43**, 070701 (2026).
11. Shuo Liu, Shao-Kai Jian\*, and **Shi-Xin Zhang\***, *Noisy Monitored Quantum Circuits*, Journal of Physics: Condensed Matter **38**, 203001 (2026).
12. Han-Ze Li, Ching Hua Lee, Shuo Liu\*, **Shi-Xin Zhang\***, and Jian-Xin Zhong\*, *Quantum Mpemba effect in long-ranged  $U(1)$ -symmetric random circuits*, Physical Review B **113**, 134310 (2026).
13. Hui Yu, Jiangping Hu\*, and **Shi-Xin Zhang\***, *Quantum Pontus-Mpemba Effects in Real and Imaginary-time Dynamics*, Physical Review B **113**, 134304 (2026).
14. Zong-Liang Li, and **Shi-Xin Zhang\***, *The Dual Role of Low-Weight Pauli Propagation: A Flawed Simulator but a Powerful Initializer for Variational Quantum Algorithms*, Physical Review Research **8**, 013266(2026).
15. Yuqi Qing, Yu-Qin Chen\*, **Shi-Xin Zhang\***, *Entanglement growth and information capacity in a quasiperiodic system with a single-particle mobility edge*, Physical Review B **113**, 064308 (2026).
16. Yu-Qin Chen\*, **Shi-Xin Zhang**<sup>†</sup>, and Shengyu Zhang\*, *Non-Markovianity benefits quantum dynamics simulation*, Advanced Quantum Technologies **9**, e00405 (2026).
17. Shuo Liu<sup>†</sup>, Hao-Kai Zhang<sup>†</sup>, Shuai Yin, **Shi-Xin Zhang\*** and Hong Yao\*, *Symmetry restoration and quantum Mpemba effects in many-body localization systems*, Science Bulletin **70**, 3991 (2025).
18. Sibao Guo, Shuai Yin, **Shi-Xin Zhang\***, and Zi-Xiang Li\*, *Skin Effect Induced Anomalous Dynamics from Charge-Fluctuating Initial States*, Physical Review B **112**, 155419 (2025).
19. Hui Yu, Zi-Xiang Li\*, and **Shi-Xin Zhang\***, *Symmetry breaking dynamics in quantum many-body systems*, Chinese Physics Letters **42**, 110602 (2025).
20. Yu-Qin Chen\* and **Shi-Xin Zhang\***, *Effective temperature in approximate quantum many-body states*, Physical Review Research **7**, 033185 (2025).
21. Shuo Liu<sup>†</sup>, Zhengzhi Wu<sup>†</sup>, **Shi-Xin Zhang\***, and Hong Yao\*, *Supersymmetry dynamics on Rydberg atom arrays*, Physical Review B **112**, L020301 (2025).
22. Hui Yu, Shuo Liu, and **Shi-Xin Zhang\***, *Quantum Mpemba effects from symmetry perspectives*, AAPPs Bulletin **35**, 17 (2025).
23. Xu Feng, Shuo Liu, **Shi-Xin Zhang\***, and Shu Chen\*, *Numerical instability of non-Hermitian Hamiltonian evolution*, Physical Review B **111**, 224310 (2025).
24. Yu-Qin Chen, Shuo Liu, and **Shi-Xin Zhang\***, *Subsystem information capacity in random circuits and Hamiltonian dynamics*, Quantum **9**, 1783 (2025).
25. **Shi-Xin Zhang**<sup>†</sup>, Jiaqi Miao<sup>†</sup>, and Chang-Yu Hsieh\*, *Variational post-selection for ground states and thermal states simulation*, Quantum Science and Technology **10**, 015028 (2025).
26. Yi-Ming Ding\*, Yan-Cheng Wang, **Shi-Xin Zhang\***, and Zheng Yan\*, *Exploring the topological sector optimization on quantum computers*, Physical Review Applied **22**, 034031 (2024).
27. Shuo Liu, Ming-Rui Li, **Shi-Xin Zhang\***, Shao-Kai Jian\*, and Hong Yao\* *Noise-induced phase transitions in hybrid quantum circuits*, Physical Review B **110**, 064323 (2024).
28. **Shi-Xin Zhang\*** and Shuai Yin\*, *Universal imaginary-time critical dynamics on a quantum computer*, Physical Review B **109**, 134309 (2024).

29. Lixue Cheng<sup>†</sup>, Yu-Qin Chen<sup>†</sup>, **Shi-Xin Zhang**<sup>\*</sup>, and Shengyu Zhang<sup>\*</sup>, *Quantum approximate optimization via learning-based adaptive optimization*, Communications Physics **7**, 83 (2024).
30. Jiaqi Miao, Chang-Yu Hsieh<sup>\*</sup>, and **Shi-Xin Zhang**<sup>\*</sup>, *Neural network encoded variational quantum algorithms*, Physical Review Applied **21**, 014053 (2024).
31. Shuo Liu, **Shi-Xin Zhang**<sup>\*</sup>, Shao-Kai Jian<sup>\*</sup>, and Hong Yao<sup>\*</sup>, *Training variational quantum algorithms with random gate activation*, Physical Review Research **5**, L032040 (2023).
32. **Shi-Xin Zhang**, Zhou-Quan Wan, Chang-Yu Hsieh<sup>\*</sup>, Hong Yao<sup>\*</sup>, and Shengyu Zhang<sup>\*</sup>, *Variational quantum-neural hybrid error mitigation*, Advanced Quantum Technologies, **6**, 2300147 (2023).
33. **Shi-Xin Zhang**<sup>†</sup>, Zhou-Quan Wan<sup>†</sup>, and Hong Yao<sup>\*</sup>, *Automatic differentiable Monte Carlo: theory and application*, Physical Review Research **5**, 033041 (2023).
34. Shuo Liu<sup>†</sup>, Ming-Rui Li<sup>†</sup>, **Shi-Xin Zhang**<sup>\*</sup>, Shao-Kai Jian<sup>\*</sup>, and Hong Yao<sup>\*</sup>, *Universal Kardar-Parisi-Zhang scaling in noisy hybrid quantum circuits*, Physical Review B **107**, L201113 (2023).
35. **Shi-Xin Zhang**, et al., *TensorCircuit: a quantum software framework for the NISQ era*, Quantum **7**, 912 (2023).
36. Shuo Liu, **Shi-Xin Zhang**<sup>\*</sup>, Chang-Yu Hsieh<sup>\*</sup>, Shengyu Zhang, and Hong Yao<sup>\*</sup>, *Probing many-body localization by excited-state variational quantum eigensolver*, Physical Review B **107**, 024204 (2023).
37. **Shi-Xin Zhang**, Chang-Yu Hsieh<sup>\*</sup>, Shengyu Zhang, and Hong Yao<sup>\*</sup>, *Differentiable quantum architecture search*, Quantum Science and Technology **7**, 045023 (2022).
38. Zhou-Quan Wan, **Shi-Xin Zhang**<sup>\*</sup>, and Hong Yao<sup>\*</sup>, *Mitigating the fermion sign problem by automatic differentiation*, Physical Review B **106**, L241109 (2022).
39. **Shi-Xin Zhang**, Chang-Yu Hsieh<sup>\*</sup>, Shengyu Zhang, and Hong Yao<sup>\*</sup>, *Neural predictor based quantum architecture search*, Machine Learning: Science and Technology **2**, 045027 (2021).
40. **Shi-Xin Zhang**, Shao-Kai Jian, and Hong Yao<sup>\*</sup>, *Quantum criticality preempted by nematicity*, Physical Review B **103**, 165129 (2021).
41. **Shi-Xin Zhang**, Shao-Kai Jian, and Hong Yao<sup>\*</sup>, *Correlated triple-Weyl semimetals with Coulomb interactions*, Physical Review B (Rapid Communication) **96**, 241111 (2017).
42. Yu-Qin Chen<sup>\*</sup> and **Shi-Xin Zhang**<sup>\*</sup>, *LLM-Driven Cross-Paradigm Design for Quantum Optimal Control*, arXiv:2607.17498 (2026).
43. Lv Zhang, **Shi-Xin Zhang**<sup>\*</sup>, Heng Fan<sup>\*</sup>, and Shuo Liu, *Revealing Entanglement-Growth Mechanisms through the Magic Barrier*, arXiv:2607.09875 (2026).
44. **Shi-Xin Zhang**<sup>\*</sup> and Yu-Qin Chen<sup>\*</sup>, *ORBIT-Q: Dual-axis benchmarking of autonomous agents in scientific quantum programming*, arXiv:2607.03105 (2026).
45. **Shi-Xin Zhang**<sup>\*</sup> and Yu-Qin Chen<sup>\*</sup>, *Quantum Subliminal Learning*, arXiv:2605.29557 (2026).
46. Hui Yu, Jiangping Hu<sup>\*</sup>, and **Shi-Xin Zhang**<sup>\*</sup>, *Post-Selection-Free Decoding of Measurement-Induced Area-Law Phases via Neural Networks*, arXiv:2604.03550 (2026).
47. **Shi-Xin Zhang**<sup>\*</sup>, Yu-Qin Chen, Weitang Li, Jiaze Sun, Wei-Guo Ma, Pei-Lin Zheng, Yu-Xiang Huang, Qi-Xiang Wang, Hui Yu, Zhuo Li, Xuyang Huang, Zong-Liang Li, Zhou-Quan Wan, Shuo Liu, Jiezhong Qiu, Jiaqi Miao, Zixuan Song, Yuxuan Yan, Kazuki Tsuoka, Pan Zhang, Lei Wang, Heng Fan, Chang-Yu Hsieh, Hong Yao, and Tao Xiang, *TensorCircuit-NG: A Universal, Composable, and Scalable Platform for Quantum Computing and Quantum Simulation*, arXiv:2602.14167 (2026).
48. Chun-Yue Zhang, **Shi-Xin Zhang**<sup>\*</sup>, and Zi-Xiang Li<sup>\*</sup>, *Bond Additivity and Persistent Geometric Imprints of Entanglement in Quantum Thermalization*, arXiv:2601.01327 (2026).
49. Xu Feng, Shuo Liu, Shu Chen<sup>\*</sup>, and **Shi-Xin Zhang**<sup>\*</sup>, *Tunable discrete quasi-time crystal from a single drive*, arXiv:2512.10303 (2025).
50. Wei-Guo Ma, Heng Fan<sup>\*</sup>, and **Shi-Xin Zhang**<sup>\*</sup>, *A Qudit-native Framework for Discrete Time Crystals*, arXiv:2512.04577 (2025).
51. Hui Yu, Jiangping Hu<sup>\*</sup>, and **Shi-Xin Zhang**<sup>\*</sup>, *Hilbert subspace imprint: a new mechanism for non-thermalization*, arXiv:2506.11922 (2025).
52. **Shi-Xin Zhang**<sup>\*</sup>, and Lei Wang<sup>\*</sup>, *A Unified Variational Framework for Quantum Excited States*, arXiv:2504.21459 (2025).
53. **Shi-Xin Zhang**, *Classification on the computational complexity of spin models*, arXiv:1911.04122 (2019).
54. Zhou-Quan Wan and **Shi-Xin Zhang**<sup>\*</sup>, *Automatic differentiation for complex valued SVD*, arXiv:1909.02659 (2019).

55. **Shi-Xin Zhang** and Hong Yao\*, *Strong and weak many-body localizations*, arXiv:1906.00971 (2019).

## Patents

**40** domestic and international patent applications (24 authorized) in the fields of quantum circuit design automation, quantum AI hybrid solutions, quantum software acceleration, and quantum simulation.

## HONORS

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- **Tsinghua University:** National Scholarship, Future Scholar Scholarship, Zhang Mingwei Scholarship, and other awards.
- **Tencent:** Rated Outstanding (top 10%); Tencent Outstanding R&D Award and Outstanding Course Award.

## SKILLS

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- **Quantum Computing:** quantum algorithms, quantum simulation, quantum noise, quantum software, quantum information, and quantum machine learning.
- **Quantum Many-Body Physics:** theory and numerical methods including tensor networks, quantum Monte Carlo, mean-field, variational methods, and exact diagonalization.
- **Generative AI, Agents, and Harnesses:** familiar with concepts and basic architectures of generative models and LLMs; agentic systems, tool use, evaluation, and benchmarking; practical use of Claude Code and Codex, and scaffolding of agent-enabled research and software-development workflows.
- **Differentiable and Scientific Computing:** TensorFlow, JAX, PyTorch, Keras, TensorNetwork, Qiskit, Cirq, TensorFlow Quantum, PennyLane, and Mitiq.
- **Engineering and HPC:** design, deployment, and operation of CPU/GPU clusters and cloud environments for large-scale scientific workloads, including toolchains, environments, and job management.
- **DevOps for Science:** reproducible research codebases through version control, automated testing and CI, packaging and release, experiment tracking, and documentation.
- **Python Ecosystem:** scientific computing, data science, machine learning, web development, and software engineering.
- **Languages:** Python, Mathematica, C++, JavaScript, and Bash; markup: HTML, CSS, Markdown, reStructuredText, and  $\LaTeX$ ; Chinese, English, and Korean.

## OPEN SOURCE CONTRIBUTIONS

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Creator and maintainer of open-source projects in quantum computing, scientific computing, HPC, and AI, with **3000+** stars/forks and 300+ followers. Contributor to NumPy, TensorFlow, Autograd, TensorNetwork, TensorFlow Quantum, and conda-smithy. Details: [GitHub](#).

## Projects

- **TensorCircuit & TensorCircuit-NG:** Full-featured, high-performance quantum software framework designed for the NISQ era. The package receives **1M+** downloads and 190+ academic citations, and has been used for research by IBM, NVIDIA, and Google. TensorCircuit was recognized among QuantumChina's Top 10 Quantum-Industry Events in 2022, recommended quantum software in Google Summer of Code 2023, and used in hackathon projects in Unitary Hack 2024 and summer-ospp 2025.
- **TensorNetwork-NG:** This is a continuously maintained fork of the popular tensor network package, originally released by Google and designed for the machine learning era.
- **qop:** The software supports algebra on complex number, quaternion, boson, hardcore boson, fermion, spin, and einsum on symbols.
- **realspace-RG:** Numerical implementation of renormalization group on many-body localization phase transition. Highly parallelized C++ code used on Tianhe II Supercomputer.
- **xalpha:** Analysis, management and backtest on financial investment. The software has **2300+** stars with a relatively large open-source community.
- **hpc-ansible:** This toolset provides all components required to build the supercomputer from bootstrap. With vm-cluster toolset, users can build a KVM cluster in one click.

- subway: The software enables HPC job management in an automatic fashion, by providing a highly customized pipeline from data management to job submission.
- myarxiv-app: The project builds the full-stack web for arXiv papers, utilizing techniques including Vue, Webpack, Flask, Docker Compose, web crawler, and NLP algorithms.